10A GENERAL PRINCIPLES IN ORGANIC CHEMISTRY

SPECIFICATION REFERENCE
10.1 10.2 10.3 10.5

LEARNING OBJECTIVES

- Be able to classify reactions as addition, elimination, substitution, oxidation, reduction, hydrolysis or polymerisation.
- Understand the concept of a reaction mechanism.
- Understand that heterolytic bond breaking results in species that are electrophiles or nucleophiles.
- Understand the link between bond polarity and the type of reaction mechanism a compound will undergo.

REACTIONS IN ORGANIC CHEMISTRY

In **Topic 4**, you learned that there were five different types of organic reaction. In this topic you will learn about two more types. All seven types of reaction are considered here so that you don't have to refer back to **Topic 4**.

ADDITION REACTIONS

In this type of reaction, two reactant species combine together to form a single product species. Usually all the species are molecules. A general equation for an addition reaction is:

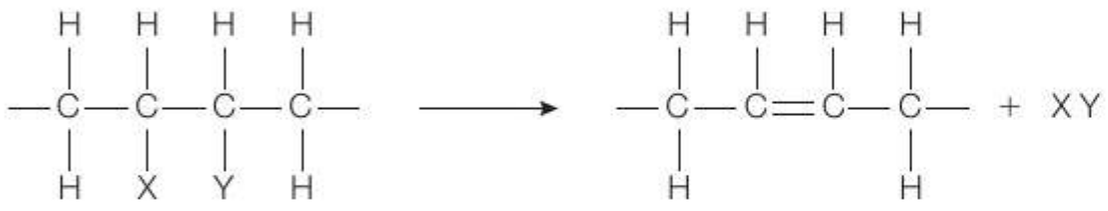
$$A + B \rightarrow C$$

One example is the reaction between ethene and bromine:

$$C_2H_4 + Br_2 \rightarrow C_2H_4Br_2$$

ELIMINATION REACTIONS

This is one of the two new types of reaction you are learning about. In this type, two atoms or groups are removed from a molecule (they are *eliminated* from the molecule). The atoms or groups are removed from neighbouring carbon atoms. This results in the formation of a second bond between the two carbon atoms. A general equation for an elimination reaction is:



X and Y represent atoms or groups of atoms. In this example, the organic product has a double bond between the central carbon atoms and a small molecule, XY, is also formed.

One example you will meet later is the reaction between bromoethane and ethanolic potassium hydroxide:

$$CH_3CH_2Br + KOH \rightarrow CH_2{=}CH_2 + KBr + H_2O$$

The small molecule eliminated is HBr (the H comes from the left-hand carbon, and the Br from the right-hand carbon). This is not easy to see just by looking at the equation. This is because the H goes to the H_2O molecule and the Br to KBr.

SUBSTITUTION REACTIONS

In this type of reaction, two reactant species combine together to form two product species. Usually all the species are molecules or ions. A general equation for a substitution reaction is:

$$A + B \rightarrow C + D$$

One example is the reaction between bromoethane and potassium hydroxide. Potassium hydroxide is an ionic compound, and as the potassium ion is a spectator ion, the reaction only involves the hydroxide ion. An equation for this reaction is:

$$C_2H_5Br + OH^- \rightarrow C_2H_5OH + Br^-$$

In this reaction, the OH group has taken the place of, or *substituted*, the Br atom. You might think that this type of reaction is like the displacement reactions you met in **Topic 8C.2**. The equation for one of these displacement reactions is $Mg + CuSO_4 \rightarrow MgSO_4 + Cu$. However, this equation can be simplified to $Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu$, which means that electrons are being transferred from Mg to Cu^{2+}, so it is a redox reaction. The reaction between bromoethane and the hydroxide ion is not a redox reaction, so it is better not to think of substitution and displacement reactions as being the same type.

OXIDATION REACTIONS

In this type of reaction, one organic compound is oxidised, usually by an inorganic reagent. This means that the organic compound can either lose hydrogen or gain oxygen. There isn't a suitable general equation that can be used for this type of reaction, but here is one example you will see later in this topic: the oxidation of ethanol by a mixture of potassium dichromate(VI) and sulfuric acid. The equation is not written to include the inorganic reagent because it would be very complicated. Usually the oxygen atoms produced by the oxidising agent are shown using the symbol [O], so the equation then becomes:

$$C_2H_5OH + [O] \rightarrow CH_3CHO + H_2O$$

You can now see why this reaction is classified as oxidation: the ethanol molecule loses two hydrogen atoms.

EXAM HINT

It is important to remember that different oxidising agents carry out different oxidation reactions. For example, acidified potassium manganate(VII) will oxidise an alkene to a diol. However, acidified potassium dichromate is needed to oxidise a primary or secondary alcohol.

REDUCTION REACTIONS

In this type of reaction, an organic compound is reduced, sometimes by hydrogen gas and a catalyst and sometimes by an inorganic reagent. This means that the organic compound can either gain hydrogen or lose oxygen. There isn't a suitable general equation that can be used for this type of reaction, but here is one example

you saw in **Topic 5A.3**: the reduction of an alkene to an alkane by hydrogen gas and a nickel catalyst. The equation for the reaction is:

$$C_2H_4 + H_2 \rightarrow C_2H_6$$

You can now see why this reaction is classified as reduction: the ethene molecule gains two hydrogen atoms. Note that this is also an example of an addition reaction.

HYDROLYSIS REACTIONS

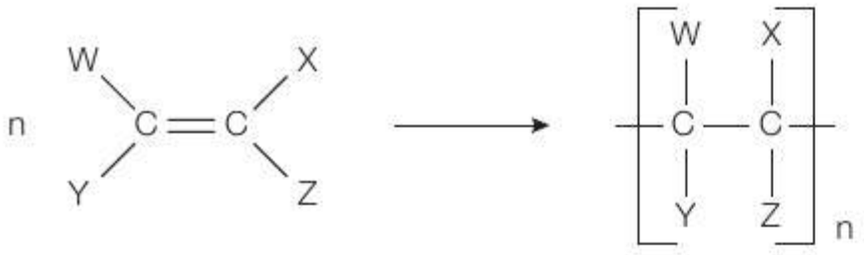
This is another new type of reaction. In hydrolysis reactions, an organic compound reacts with water. The OH group of water replaces an atom or group in the organic compound. A general equation for a hydrolysis reaction is:

$$RX + H_2O \rightarrow ROH + HX$$

You might think that this is an example of a substitution reaction, and you would be right! Hydrolysis reactions are really substitution reactions in which the OH from a water molecule replaces an atom or group (usually a halogen) in an organic compound.

POLYMERISATION REACTIONS

At IAS, all the polymerisation reactions you see are examples of addition polymerisation. If you study chemistry in more detail at IAL, you will also meet examples of condensation polymerisation. However, we will not consider these in this topic. In addition polymerisation, very large numbers of a reactant molecule (sometimes of two different reactant molecules) react together to form one very large product molecule. A general equation for a polymerisation reaction is:



A familiar example of this type of reaction is the polymerisation of ethene to poly(ethene).

REACTION MECHANISMS

You have already learnt about reaction mechanisms in **Topics 4** and **5**. Just to remind you, a mechanism tries to explain the actual changes that occur during a reaction, especially in the bonding between the atoms. A mechanism is a sequence of two or more steps, each one represented by an equation, that shows how a reaction takes place.

So far in this book, you have met two different types of reaction mechanism:

Mechanism 1 free radical substitution in alkanes

Mechanism 2 electrophilic addition to alkenes

In this topic, you will look at one more:

Mechanism 3 nucleophilic substitution in halogenoalkanes

HOMOLYTIC AND HETEROLYTIC BOND BREAKING

In mechanism 1, the type of bond breaking is homolytic. This is when a covalent bond breaks, and each atom keeps one electron from the shared pair of electrons in the bond. Free radicals are formed.

In mechanisms 2 and 3, the type of bond breaking is heterolytic, when a covalent bond breaks, and one atom keeps both electrons from the shared pair of electrons in the bond. This atom becomes a negative ion, and can act as a nucleophile. The other atom becomes a positive ion, and can act as an electrophile.

ELECTROPHILES AND NUCLEOPHILES

In mechanism 1, the attacking species are free radicals, such as Cl•.

In mechanism 2, the attacking species are electrophiles, such as the H of an HBr molecule.

In mechanism 3, the attacking species are nucleophiles, such as the OH^- ion.

THE ROLE OF BOND POLARITY IN MECHANISMS

In mechanism 1, the reactants are alkanes and halogens, whose bonds are either non-polar or only very slightly polar. This means that the type of bond breaking is most likely to be homolytic.

In mechanisms 2 and 3, the reactants are molecules such as hydrogen halides and halogenoalkanes, whose bonds are polar. This means that the type of bond breaking is most likely to be heterolytic.

LEARNING TIP

You can often work out the type of an unfamiliar reaction by looking at its equation. Also, some reactions can be classified as more than one type, such as addition and reduction, if they both involve hydrogen reacting with an organic reactant to form a single organic product.

CHECKPOINT

1. The equations for some reactions are given below. Classify each reaction, by looking at the equations.

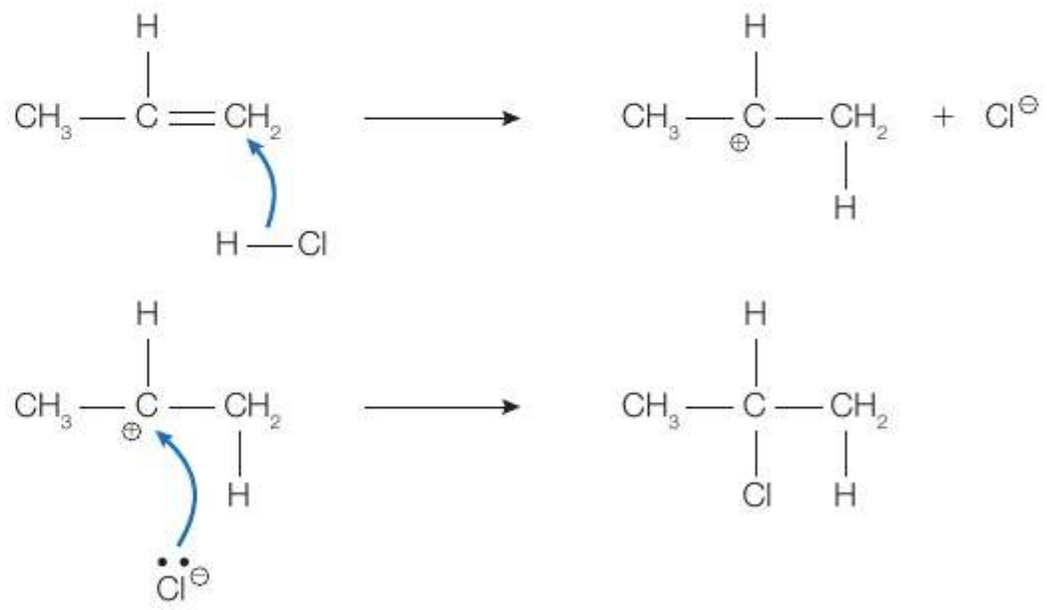
(a) $CH_3CHClCH_3 + LiOH \rightarrow CH_3CH(OH)CH_3 + LiCl$

(b) $CH_3CH{=}CH_2 + H_2O \rightarrow CH_3CH(OH)CH_3$

(c) $CH_3CH_2CHO + [O] \rightarrow CH_3CH_2COOH$

(d) $CH_3CHClCH_3 + NaOH \rightarrow CH_3CH{=}CH_2 + NaCl + H_2O$

2. A student wrote a mechanism for the reaction between propene and hydrogen chloride:



Identify the mistakes in the student's mechanism.